

Uber on the Cloud

Architecture & Case Study Analysis

ENGINEERING A CLOUD Formative Assessment 1 (Units 1–4 Mapping)

Third Year B.Tech · Cloud Computing

ATHARVA VIKHE - 124B1F127

OM SHETE - 124B1F135

OBJECTIVE

Evaluating Uber's real-world cloud infrastructure mapped against academic core units

Uber's Engineering Challenge & Scale

Operating across diverse geographies, Uber manages millions of real-time transactions daily, ranging from ride-sharing to food delivery. This global scale demands an exceptionally robust and highly scalable cloud infrastructure capable of dynamic resource allocation and real-time data processing.

The platform's complexity extends to supporting a vast ecosystem of interdependent microservices, handling petabytes of data, and maintaining ultra-low latency for critical operations like driver-rider matching, dynamic pricing, and navigation across more than 70 countries.



Uber's Engineering Challenge & Scale

70+

Countries

Active global footprint

<1s

Latency Target

Sub-second driver-rider matching

99.99%

Uptime SLA

Across all global regions

From legacy on-premise data centres to a resilient **Multi-Cloud architecture** — GCP & AWS — built for real-time surge pricing, routing, and dispatch.



Deployment & Service Model Breakdown



Public Multi-Cloud Deployment

GCP primary · AWS auxiliary — eliminates vendor lock-in and single-point failure.

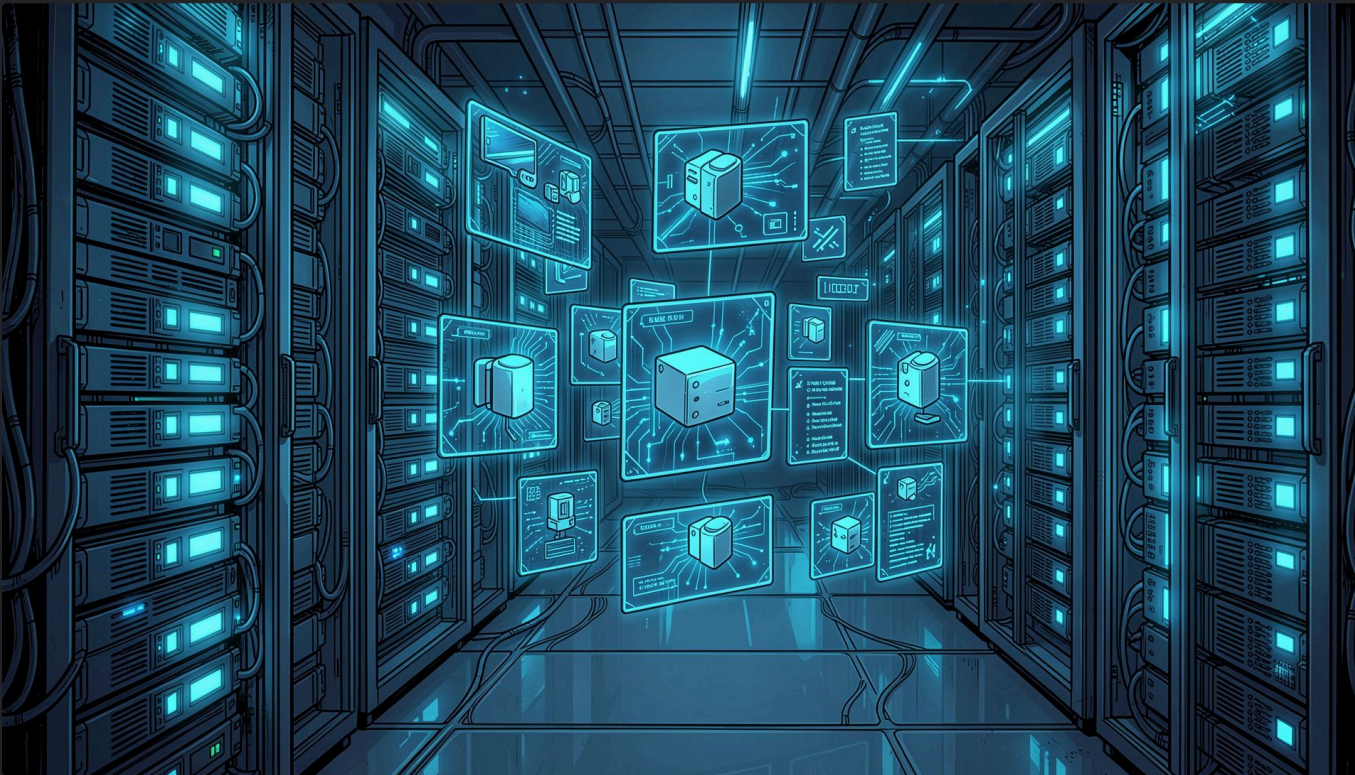
IaaS

Compute Engine instances for core server processing

PaaS

Managed Kubernetes, AI/ML pipelines, database engines

Virtualisation & Compute Workloads



Google Compute Engine VMs

Custom VM clusters that auto-scale during peak demand — weekend rush, holidays

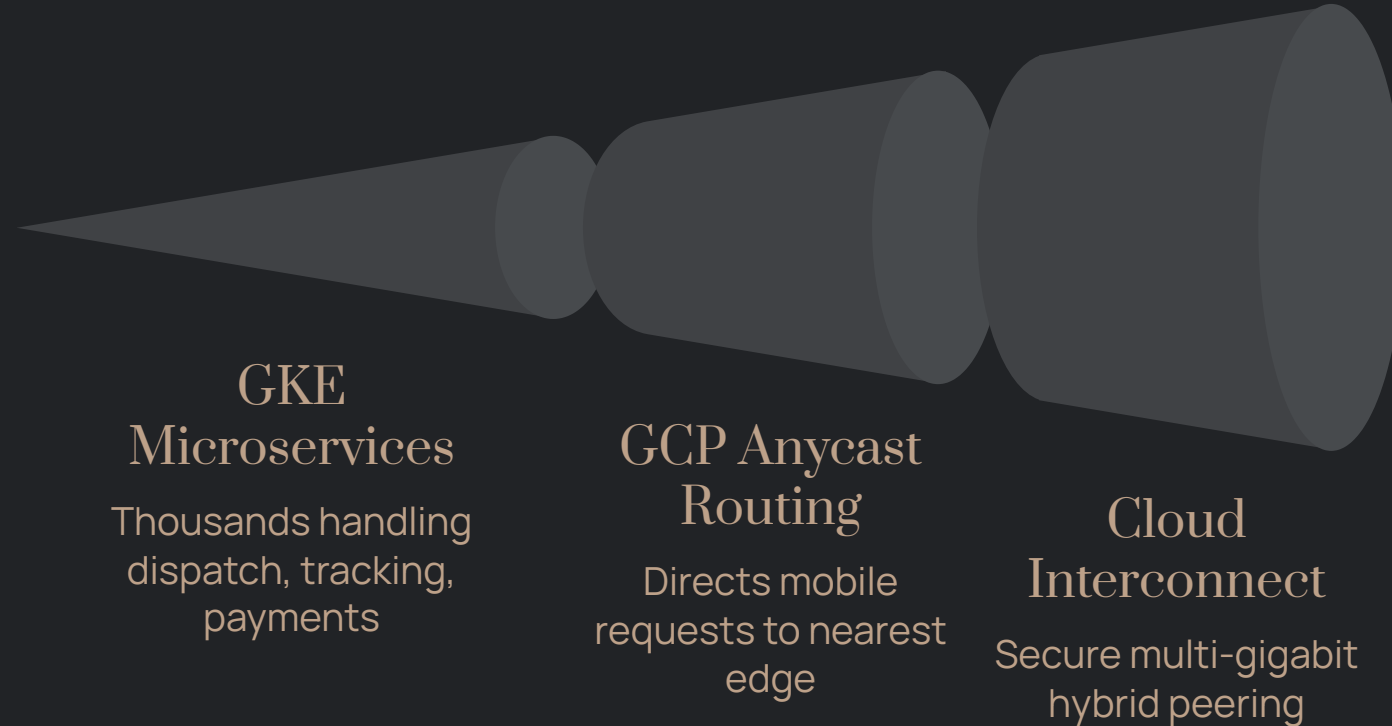
Serverless Functions

Event-driven processing for push notifications, trip receipts, driver status updates

Dynamic Capacity Allocation

Ensures 99.99% uptime across all global regions

Containers & Global Networking



GKE orchestrates microservices; Anycast routing ensures sub-second dispatch; Cloud Interconnect secures multi-cloud connectivity.

Cloud Security Architecture



IAM & RBAC

Strict Role-Based Access Control with least-privilege policies enforced across all cloud services



Threat Mitigation

Google Cloud Armor — enterprise-grade DDoS defence and WAF rule filtering



Encryption & Secrets

TLS 1.3 in transit · AES-256 at rest · Keys managed via GCP KMS & Secret Manager



Compliance & Data Privacy Standards



PCI-DSS Level 1

Full compliance for millions of global card and digital wallet transactions



GDPR & CCPA

Driver/passenger location logs and PII handled per EU and US data privacy laws

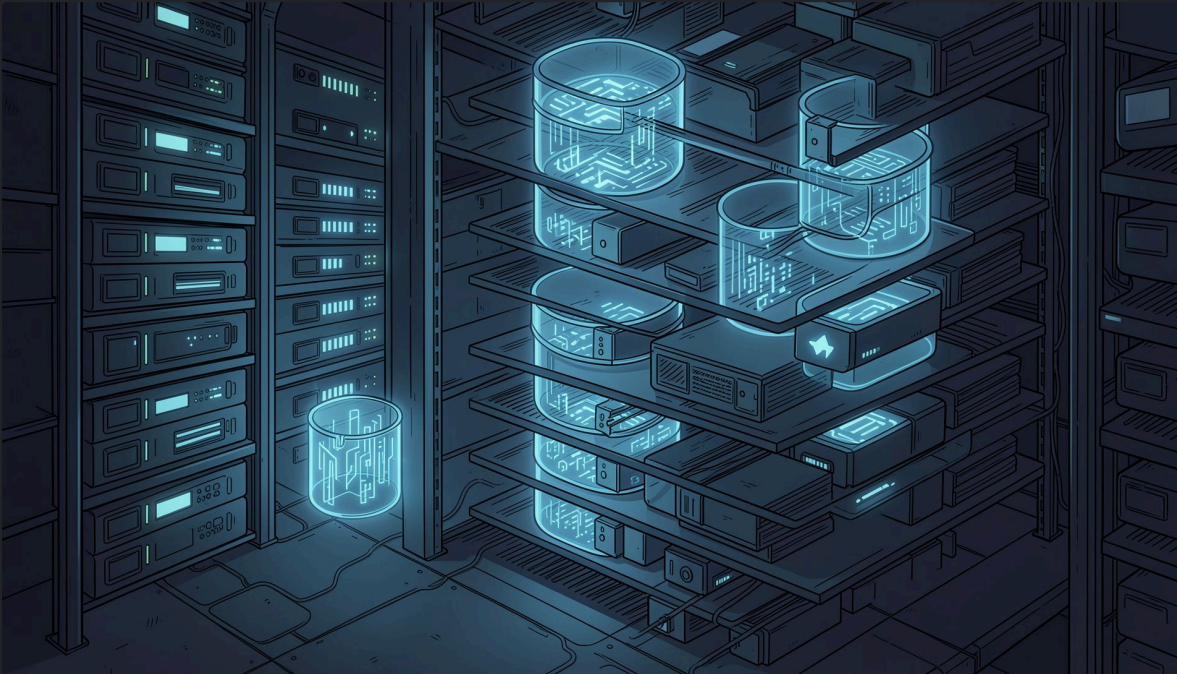


Auditing & Governance

Cloud Audit Logs backed by SOC 1, SOC 2, and ISO 27001 certifications



Cloud Storage Infrastructure



Google Cloud Storage (GCS)

Object storage for driver IDs, profile photos, ride receipts, and map tiles.

Standard Storage

High-speed access for active trip data and live user media

Coldline / Archive

Cost-effective long-term archival for historical trip logs and financial audit trails

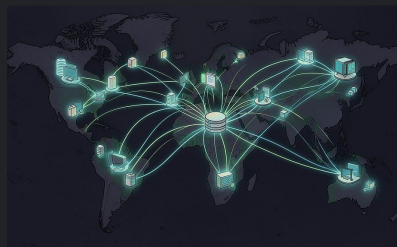
UNIT 4

Database Systems & Scale



Bigtable — NoSQL

Real-time processing of millions of driver
GPS pings per second



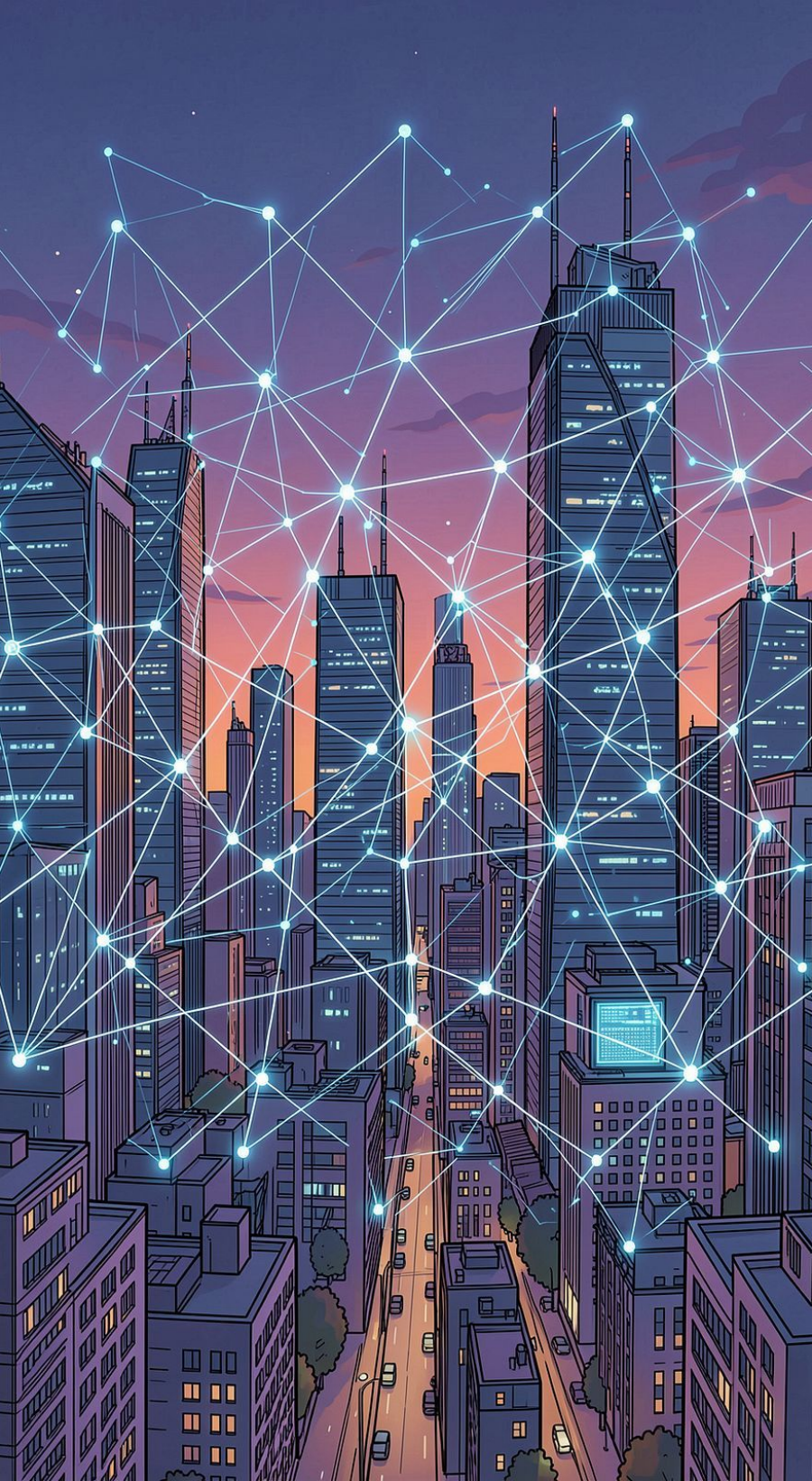
Cloud Spanner — Relational

Globally consistent, ACID-compliant
ledger and billing records



BigQuery — Analytics

Exabyte-scale data warehouse
optimising fare algorithms and driver
distribution



Summary & Key Takeaways

01

Unit 1 — Deployment

Multi-cloud IaaS/PaaS delivers flexibility and high availability

02

Unit 2 — Compute & Networking

GKE and Anycast routing enable real-time sub-second dispatch

03

Unit 3 — Security

IAM, KMS, and PCI-DSS safeguard global financial and GPS data

04

Unit 4 — Storage & Data

Bigtable and Cloud Spanner handle Exabyte-scale workloads



Thank you — open for questions and discussion.